Lab 6: Dynamic Routing Configurations: RIP, EIGRP, OSPF, BGP

Objectives

- To grasp how routers automatically discover remote networks and maintain routing tables without manual entry of every path.
- To configure and analyses the operation of RIP, EIGRP, OSPF, and BGP, focusing on how different algorithms (Distance Vector, Link-State, and Path Vector) automatically discover network topologies and populate routing tables.

Theory

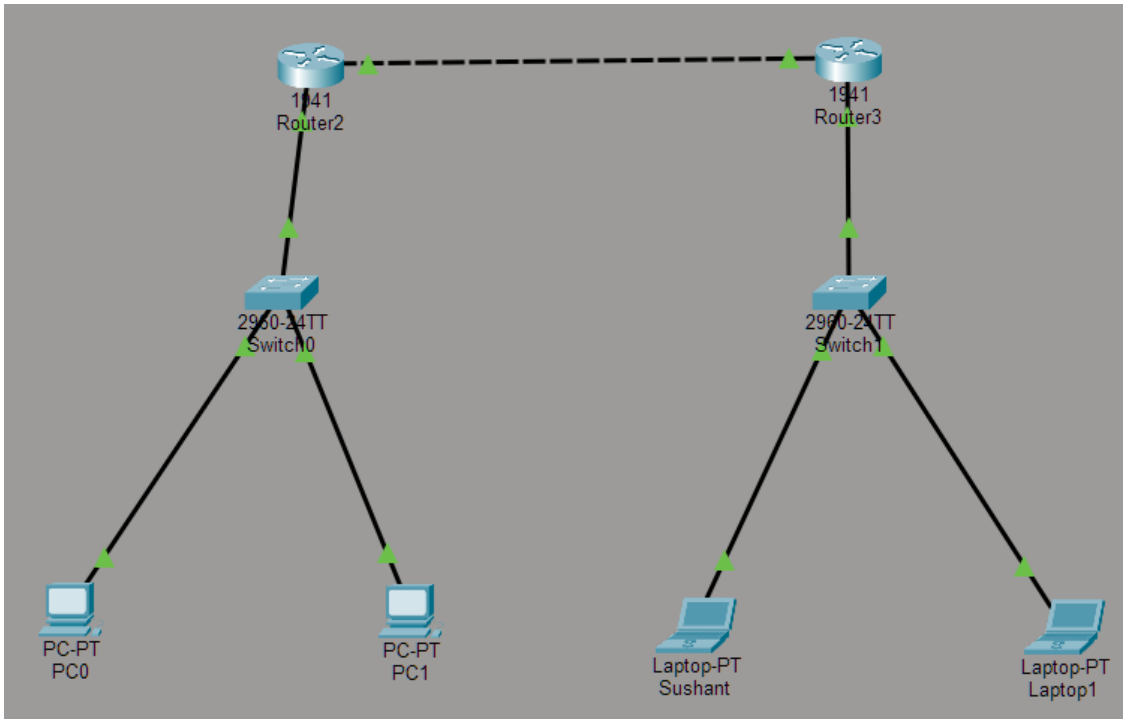
Dynamic routing protocols allow routers to automatically discover remote networks and build their own routing tables. Unlike static routing, these protocols can adapt to topology changes without manual intervention.

1. Routing Information Protocol (RIP)

RIP is one of the oldest distance vector protocols. It is characterized by its simplicity but has limitations in larger networks.

- Algorithm: Uses the Distance Vector algorithm.
- Metric: Uses Hop Count to determine the best path, with a maximum limit of 15 hops.
- Operation: Routers send their entire routing table to their neighbors every 30 seconds.
- Versions: RIP v1 is classful (no subnetting support), while RIP v2 is classless and supports CIDR.

Network Topology



Configuration

For PCs:

Device	IPv4 Address	Subnet Mask	Default Gateway
PC0	192.168.1.2	255.255.255.0	192.168.1.1
PC1	192.168.1.3	255.255.255.0	192.168.1.1
PC2	192.168.2.2	255.255.255.0	192.168.2.1
PC3	192.168.2.3	255.255.255.0	192.168.2.1

For Routers

Device	IPv4 Address	Subnet Mask	Next Hop	Ethernet Cable	Default Gateway
Router 1	192.168.2.0	255.255.255.0	10.0.0.2	Gig 0/0	10.0.0.1
Router 1				Gig 0/1	192.168.1.1
Router 2	192.168.1.0	255.255.255.0	10.0.0.1	Gig 0/0	10.0.0.2
Router 2				Gig 0/1	192.168.2.1

Commands:

Router 0	Router 1
router rip version 2 network 192.168.1.0 network 10.0.0.0 no auto-summary	router rip version 2 network 192.168.2.0 network 10.0.0.0 no auto-summary

Observation

After enabling RIP v2 on both routers, a ping was performed from PC0 (192.168.1.2) to PC3 (192.168.2.3):

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.2.3

Pinging 192.168.2.3 with 32 bytes of data:

Request timed out.
Reply from 192.168.2.3: bytes=32 time<1ms TTL=126
Reply from 192.168.2.3: bytes=32 time<1ms TTL=126
Reply from 192.168.2.3: bytes=32 time<1ms TTL=126

Ping statistics for 192.168.2.3:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
C:\>ping 192.168.2.3

Reply from 192.168.2.3: bytes=32 time<1ms TTL=126
Reply from 192.168.2.3: bytes=32 time<1ms TTL=126
```

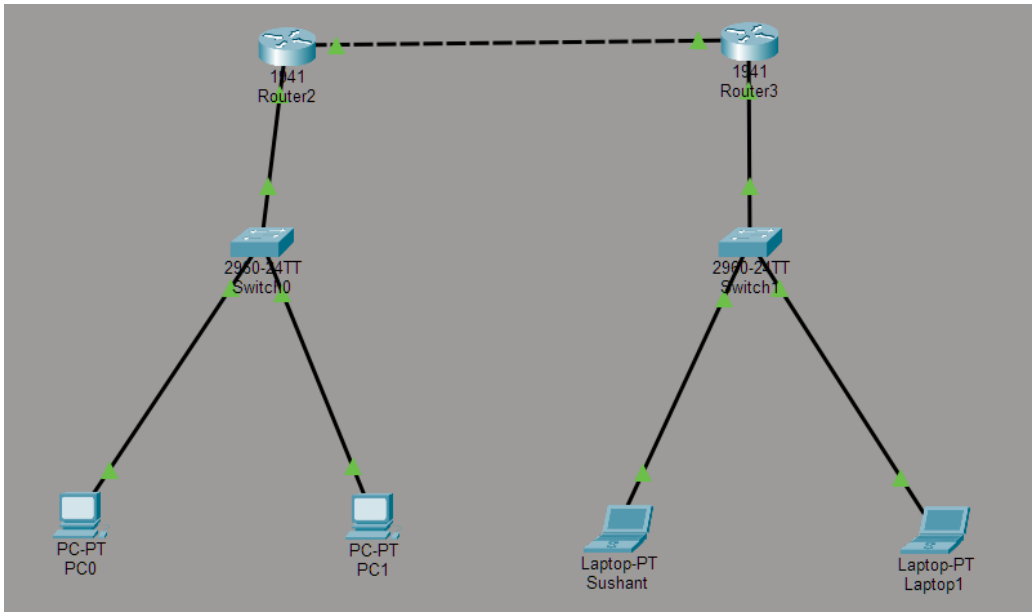
```
Reply from 192.168.2.3: bytes=32 time<1ms TTL=126
Reply from 192.168.2.3: bytes=32 time<1ms TTL=126

Ping statistics for 192.168.2.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
C:\>
```

2. Enhanced Interior Gateway Routing Protocol (EIGRP)

EIGRP is a hybrid protocol that combines the best features of both distance vector and link-state protocols.

- Algorithm: Uses the Diffusing Update Algorithm (DUAL) to ensure loop-free paths.
- Metric: Uses a composite metric based on Bandwidth and Delay (optionally Reliability and Load).
- Operation: Routers only send updates when a topology change occurs, which saves bandwidth.
- Feature: Supports unequal cost load balancing, which is unique among these protocols.

Network Topology



Configuration

For PCs:

Device	IPv4 Address	Subnet Mask	Default Gateway
PC0	192.168.1.2	255.255.255.0	192.168.1.1
PC1	192.168.1.3	255.255.255.0	192.168.1.1
PC2	192.168.2.2	255.255.255.0	192.168.2.1
PC3	192.168.2.3	255.255.255.0	192.168.2.1

For Routers:

Device	IPv4 Address	Subnet Mask	Next Hop	Ethernet Cable	Default Gateway
Router 1	192.168.2.0	255.255.255.0	10.0.0.2	Gig 0/0	10.0.0.1
Router 1				Gig 0/1	192.168.1.1
Router 2	192.168.1.0	255.255.255.0	10.0.0.1	Gig 0/0	10.0.0.2
Router 2				Gig 0/1	192.168.2.1

Commands:

Router 0	Router 1
router eigrp 100 network 192.168.1.0 0.0.0.255 network 10.0.0.0 0.0.0.255 no auto-summary	router eigrp 100 network 192.168.2.0 0.0.0.255 network 10.0.0.0 0.0.0.255 no auto-summary

Observation

After configuring EIGRP on both routers, a ping was performed from PC0 to PC2 (192.168.2.2):

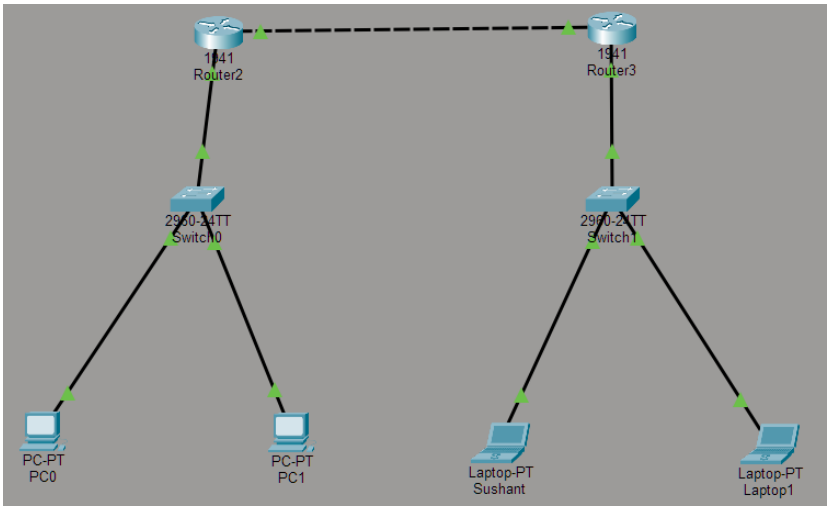
```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.2.2

Pinging 192.168.2.2 with 32 bytes of data:

Request timed out.
Reply from 192.168.2.2: bytes=32 time<1ms TTL=126
Reply from 192.168.2.2: bytes=32 time<1ms TTL=126
Reply from 192.168.2.2: bytes=32 time<1ms TTL=126

Ping statistics for 192.168.2.2:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
C:\>ping 192.168.2.2

Reply from 192.168.2.2: bytes=32 time<1ms TTL=126
Reply from 192.168.2.2: bytes=32 time<1ms TTL=126
Reply from 192.168.2.2: bytes=32 time<1ms TTL=126
Reply from 192.168.2.2: bytes=32 time<1ms TTL=126

Ping statistics for 192.168.2.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
C:\>
```

3. Open Shortest Path First (OSPF)

OSPF is a widely used link-state routing protocol designed for scalability and fast convergence.

- Algorithm: Uses Dijkstra's Shortest Path First (SPF) algorithm.
- Metric: Uses Cost, which is calculated based on reference bandwidth divided by interface bandwidth.
- Operation: Routers discover neighbors using "Hello" packets and exchange Link State Advertisements (LSAs) to build a synchronized topology database.
- Advantage: Offers very fast convergence compared to RIP.

Network Topology



Configuration

For PCs:

Device	IPv4 Address	Subnet Mask	Default Gateway
PC0	192.168.1.2	255.255.255.0	192.168.1.1
PC1	192.168.1.3	255.255.255.0	192.168.1.1
PC2	192.168.2.2	255.255.255.0	192.168.2.1
PC3	192.168.2.3	255.255.255.0	192.168.2.1

For Routers:

Device	IPv4 Address	Subnet Mask	Next Hop	Ethernet Cable	Default Gateway
Router 1	192.168.2.0	255.255.255.0	10.0.0.2	Gig 0/0	10.0.0.1
Router 1				Gig 0/1	192.168.1.1
Router 2	192.168.1.0	255.255.255.0	10.0.0.1	Gig 0/0	10.0.0.2
Router 2				Gig 0/1	192.168.2.1

Commands:

Router 0	Router 1
router ospf 1 network 192.168.1.0 0.0.0.255 area 0 network 10.0.0.0 0.0.0.255 area 0	router ospf 1 network 192.168.2.0 0.0.0.255 area 0 network 10.0.0.0 0.0.0.255 area 0

Observation

After configuring OSPF on both routers, a ping was performed from PC0 to PC2 (192.168.2.2):

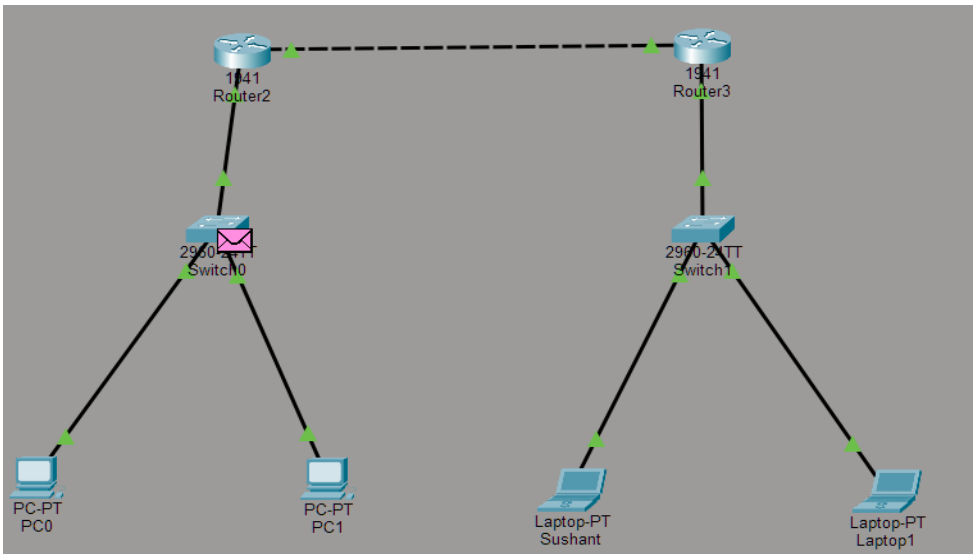
```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.2.2

Pinging 192.168.2.2 with 32 bytes of data:

Request timed out.
Reply from 192.168.2.2: bytes=32 time<1ms TTL=126
Reply from 192.168.2.2: bytes=32 time<1ms TTL=126
Reply from 192.168.2.2: bytes=32 time<1ms TTL=126

Ping statistics for 192.168.2.2:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
C:\>ping 192.168.2.2

Reply from 192.168.2.2: bytes=32 time<1ms TTL=126
Reply from 192.168.2.2: bytes=32 time<1ms TTL=126
Reply from 192.168.2.2: bytes=32 time<1ms TTL=126
Reply from 192.168.2.2: bytes=32 time<1ms TTL=126

Ping statistics for 192.168.2.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Minimum = 0ms, Maximum = 1ms, Average = 0ms
C:\>
```

4. Border Gateway Protocol (BGP)

BGP is a path vector protocol and is considered the "routing protocol of the internet".

- Purpose: It is used to connect different Autonomous Systems (AS), which are groups of networks under a single administration.
- Operation: BGP uses path attributes (like AS-Path) rather than simple metrics to select the best path.
- Reliability: It uses TCP port 179 for reliable communication between routers.
- Scalability: It is highly reliable and scales to manage the entire global internet routing table.

Network Topology



Configuration

For PCs:

Device	IPv4 Address	Subnet Mask	Default Gateway
PC0	192.168.1.2	255.255.255.0	192.168.1.1
PC1	192.168.1.3	255.255.255.0	192.168.1.1
Laptop-PT Sushant	192.168.2.2	255.255.255.0	192.168.2.1
Laptop-PT laptop01	192.168.2.3	255.255.255.0	192.168.2.1

For Routers:

Device	IPv4 Address	Subnet Mask	Next Hop	Ethernet Cable	Default Gateway
Router 1	192.168.2.0	255.255.255.0	10.0.0.2	Gig 0/0	10.0.0.1
Router 1				Gig 0/1	192.168.1.1
Router 2	192.168.1.0	255.255.255.0	10.0.0.1	Gig 0/0	10.0.0.2
Router 2				Gig 0/1	192.168.2.1

Commands:

Assumption: R1 → AS 65001, R2 → AS 65002

Router 0	Router 1
router bgp 65001 neighbor 10.0.0.2 remote-as 65002 network 192.168.1.0 mask 255.255.255.0	router bgp 65002 neighbor 10.0.0.1 remote-as 65001 network 192.168.2.0 mask 255.255.255.0

Observation

After configuring BGP, a ping was performed from PC2 to PC1 (192.168.1.3):

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.3

Pinging 192.168.1.3 with 32 bytes of data:

Request timed out.
Reply from 192.168.1.3: bytes=32 time<1ms TTL=126
Reply from 192.168.1.3: bytes=32 time<1ms TTL=126
Reply from 192.168.1.3: bytes=32 time<1ms TTL=126

Ping statistics for 192.168.1.3:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
C:\>ping 192.168.1.3

Reply from 192.168.1.3: bytes=32 time<1ms TTL=126
Reply from 192.168.1.3: bytes=32 time<1ms TTL=126
Reply from 192.168.1.3: bytes=32 time<1ms TTL=126
Reply from 192.168.1.3: bytes=32 time<1ms TTL=126

Ping statistics for 192.168.1.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
C:\>
```

Result

In this lab, we moved away from manual routing and successfully set up four different dynamic protocols: RIP, EIGRP, OSPF, and BGP. For each configuration, the routers were able to automatically exchange network information without manual path entry. We verified this by checking the routing tables and confirming successful ping connectivity across the entire network

Discussion

In this lab, we implemented different dynamic routing configurations and tested each one to differentiate it from the other. RIP was the simplest to set up, but it is a bit basic since it only cares about how many routers are in the way. OSPF felt much more professional because it looks at link speed bandwidth to calculate the cost of a path. EIGRP was the fastest to react when topology changes were made. BGP showed us how the big internet works by grouping networks into Autonomous Systems rather than just looking at individual routers. Ping commands were run after each configuration to confirm connectivity.

Conclusion

In this lab, various algorithms of dynamic routing were successfully implemented using Cisco Packet Tracer. Each protocol RIP, EIGRP, OSPF, and BGP demonstrated its unique approach to automatic route discovery and path selection, confirming that dynamic routing is more scalable and adaptive compared to static routing.